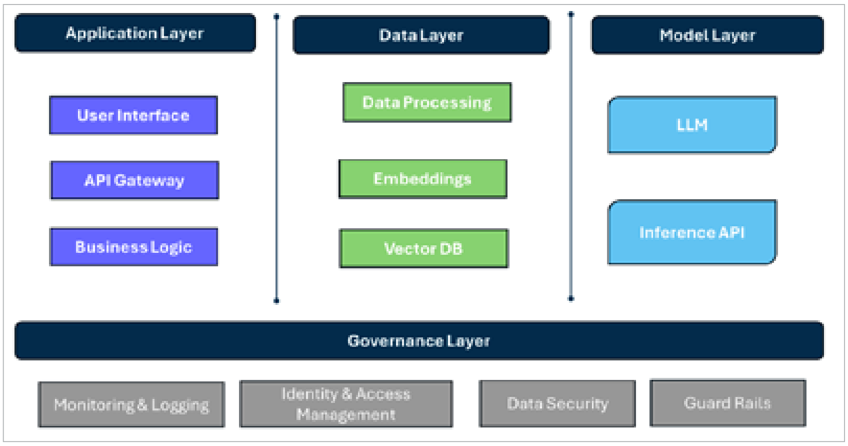

Figure 1: Generative AI application building blocks

leveraging graph databases such as Neo4J for knowledge graphs.

Model layer: This layer takes care of model development, training, deployment and serving. As model training is an expensive task, teams are leveraging commercial models offered by major cloud providers such as Azure, OpenAI, AWS, and Google AI/ML LLM models. Open source frameworks such as Ollama are available for local deployment. HuggingFace offers model deployment services.

Governance layer: This layer includes enterprise essentials such as controlled and secured access, data protection, logging and monitoring. Guardrails and other features help comply with enterprise AI policies.

The role of containers in AI deployment

Many of these layers are isolated modules. GenAI application plugs them together to form a cohesive solution. We can leverage containerisation technology to effectively deploy and manage GenAI solutions.

Containers are compact, portable units that bundle an application along with its dependencies, ensuring uniformity across diverse environments. Docker is a well-known containerisation platform that eases the process of creating, deploying, and managing containers. While Docker is adept at handling individual containers, Kubernetes is an orchestration tool designed to manage containerised applications at a larger scale.

For generative AI applications, containers offer several benefits.

Isolation: Containers isolate the application from the host system, preventing conflicts between dependencies and ensuring a consistent runtime environment.

Portability and consistency: Containers can run on any system that supports containerisation, making it easy to move applications between development, testing, and production environments. They provide a consistent environment across different stages of the development lifecycle. By containerising AI/ML applications, developers can ensure that the code runs identically in development, testing, and production environments. This portability simplifies collaboration among teams and reduces the "it works on my machine" syndrome.

Scalability and resource management: GenAI and machine learning workloads can be highly variable, requiring extensive computational resources for training models. Containers excel in scaling up or down to meet the demands of the application, ensuring optimal performance. This dynamic scaling is particularly beneficial for handling large datasets and complex models typical in AI/ML applications.

Automated deployment and updates: Continuous integration and continuous deployment (CI/CD) pipelines are integral to modern AI/ML workflows. Kubernetes supports automated deployments and rollbacks, enabling teams to iterate quickly and deploy new models without downtime. This automation enhances the reliability and robustness of AI/ML services.

Resource optimisation: Kubernetes efficiently manages computational resources, allocating CPU, memory, and GPU resources as needed. For AI/ML applications that often require significant GPU resources, Kubernetes can schedule and optimise the use of available GPUs, ensuring that expensive hardware is utilised effectively.

Load balancing: Kubernetes automatically distributes network traffic across containers, ensuring high availability and reliability.

Self-healing: Kubernetes can automatically restart failed containers, replace unresponsive nodes, and reschedule containers to maintain the desired state of the application.

Data management: AI/ML applications are data-intensive, and managing data storage and access is crucial. Kubernetes supports various storage solutions, from local storage to cloud-based options, allowing seamless integration with data lakes and databases. Persistent storage in Kubernetes ensures that data remains available across pod restarts and crashes.

Security and compliance: Kubernetes offers robust security features, including role-based access control (RBAC), network policies, and secrets management. These features help teams enforce security best practices and comply with regulations, which is particularly important in industries like healthcare and finance where AI/ML applications handle sensitive data.